MAJOR PROJECT · 7TH SEMESTER

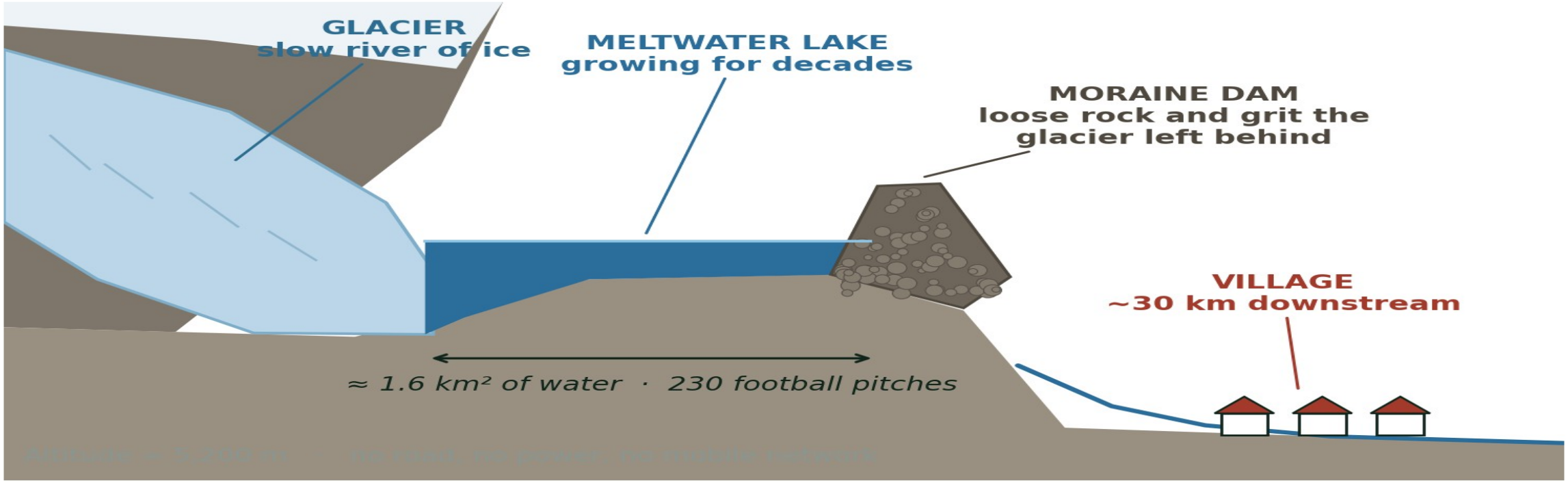
GEWS — Glacial Lake Outburst Flood Early-Warning System

Watching 41 dangerous Himalayan lakes from space, with free satellite data.
The problem, the method, the technology, the results and the limits.

Dept. of Artificial Intelligence & Machine Learning, DSCE

1 The problem

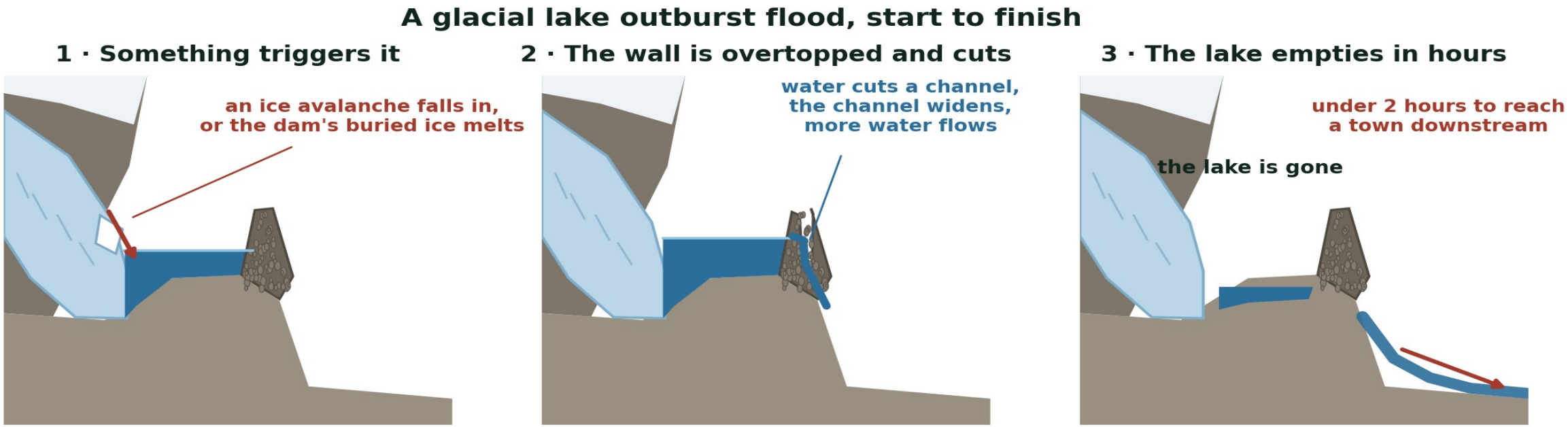
A lake held back by a wall of loose rubble that nobody built and nobody inspects



A moraine dam is gravel, boulders and often buried ice. South Lhonak held about 1.6 km² of water at 5,200 m, with villages 30 km downstream.

2 What a GLOF is

Glacial Lake Outburst Flood — the wall fails and the lake leaves all at once



Under two hours from failure to a town downstream. No detect-decide-warn-evacuate chain fits in that window — so a useful warning has to come weeks or months earlier, which means watching the lake, not the flood.

3 Why satellites, and not sensors on the lake

The obvious answer, and why it does not survive the Himalaya

The obvious idea	Why it fails here
Water-level sensor in the lake	4,500–5,300 m, no road. A multi-day expedition per lake, and there are thousands.
Camera or weather station	No mains power, no mobile network, −25 °C winters, months under snow.
Anything wired to the dam	Destroyed by the exact event it was installed to measure, and silent about the months before.
Regular field inspection	One survey per lake per year at best. A lake can change materially between visits.

So we use the instrument already up there, already paid for, already looking at every one of these lakes every few days: **Earth-observation satellites. Total data cost of this project: zero.**

4 What GEWS is — and is not

The scope is deliberately narrow, and the narrowness is a safety decision

IT IS

- Decision support for a trained disaster-management officer.
- A ranking of 41 fixed lakes by how far each has drifted from its own multi-year normal.
- A weeks-to-months indicator, not minutes-to-hours.
- Honest about blindness: where it cannot see, it says "cannot assess".

IT IS NOT

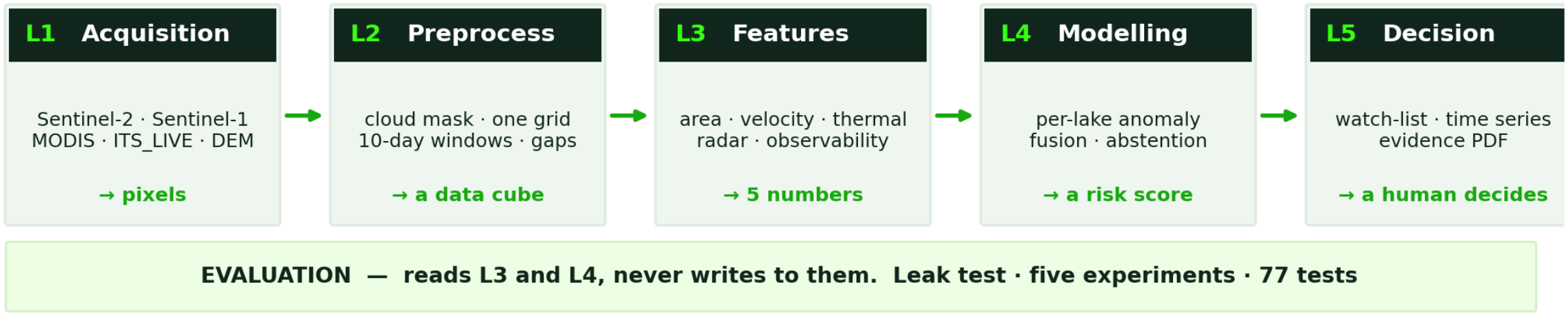
- A public alerting system. It issues no alerts, under any configuration.
- A real-time flood warning. The time step is 10 days.
- A flood model — nothing about depth or reach downstream.
- A guarantee of safety. Ever.

"Where the system cannot see, it must say so."

From the requirements specification — and the line the whole design is built around.

5 The system on one line

Five layers, one command rebuilds every number from scratch



Each layer writes a file with a declared schema and the next reads it. **That makes any layer testable on its own, lets four students work in parallel, and traces a wrong number to exactly one layer.**

6 The five data sources

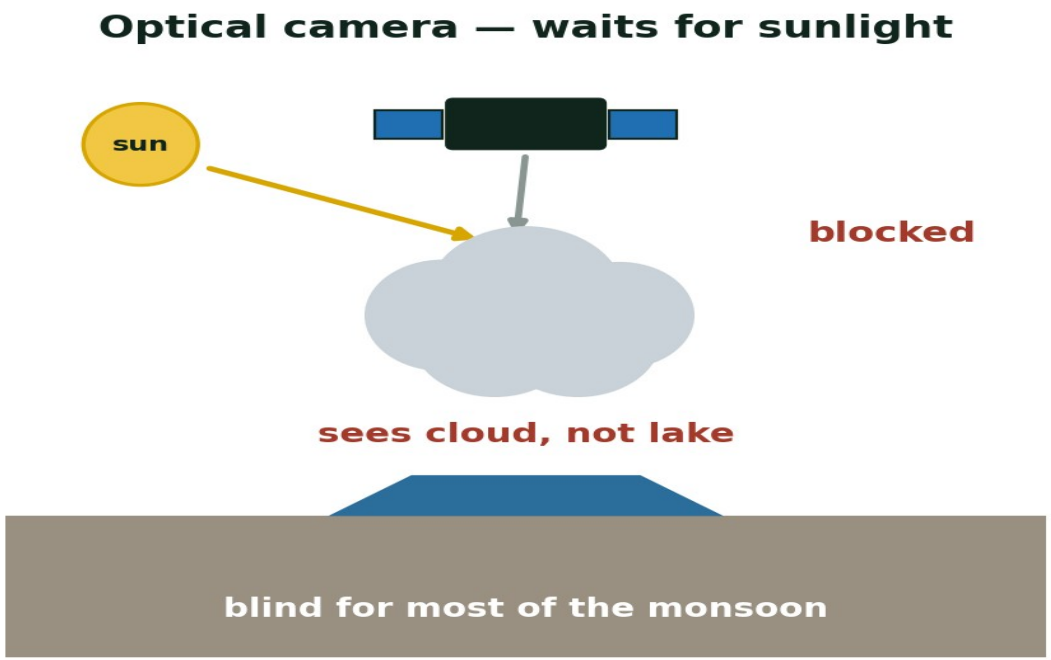
All free, all openly licensed, none needing an account

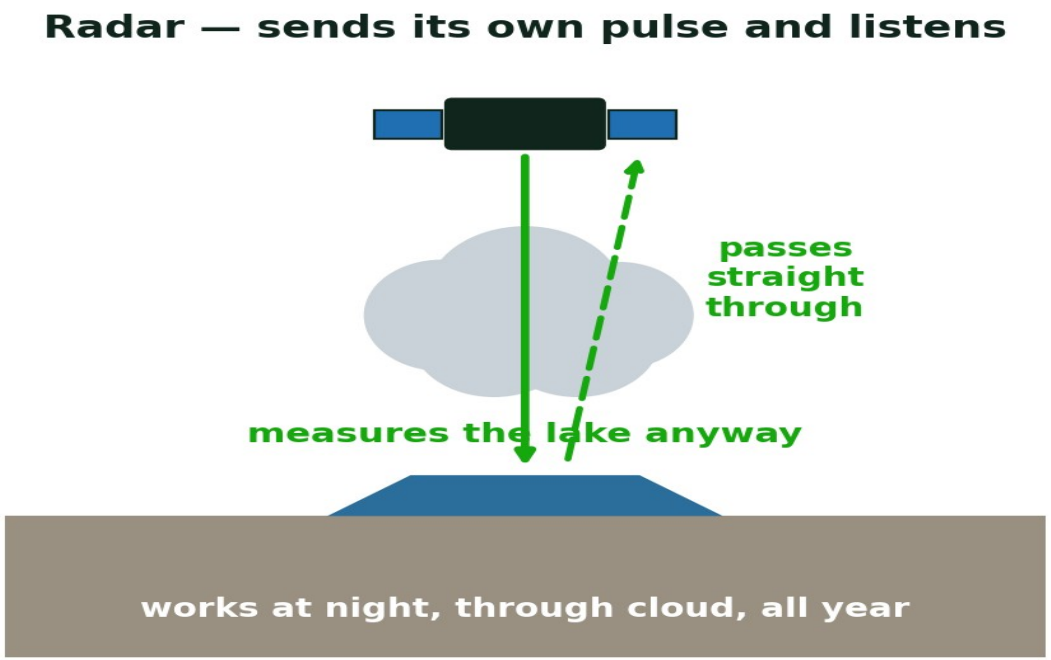
Source	In plain words	Gives	Why it is here
Sentinel-2 (ESA)	A colour camera in space	10 m optical → lake area	The primary signal: is the lake growing?
Sentinel-1 (ESA)	Radar that sees through cloud	Backscatter in dB	The monsoon fix — optical goes blind, radar does not
ITS_LIVE (NASA)	A speedometer for the glacier	Ice velocity, m/yr	Glacier acceleration is a documented precursor
MODIS (NASA)	A thermometer	Land surface temp → °C-days	How hard the melt season is pushing
Copernicus DEM	A 3-D map of the ground	Elevation and slope, 30 m	The geometry of the moraine dam. Static, read once.

Four sensors is not showing off. A single-sensor system is blind for most of the monsoon — exactly when these lakes are most active.

7 Why radar is in the system

The single idea that explains the project's strongest result

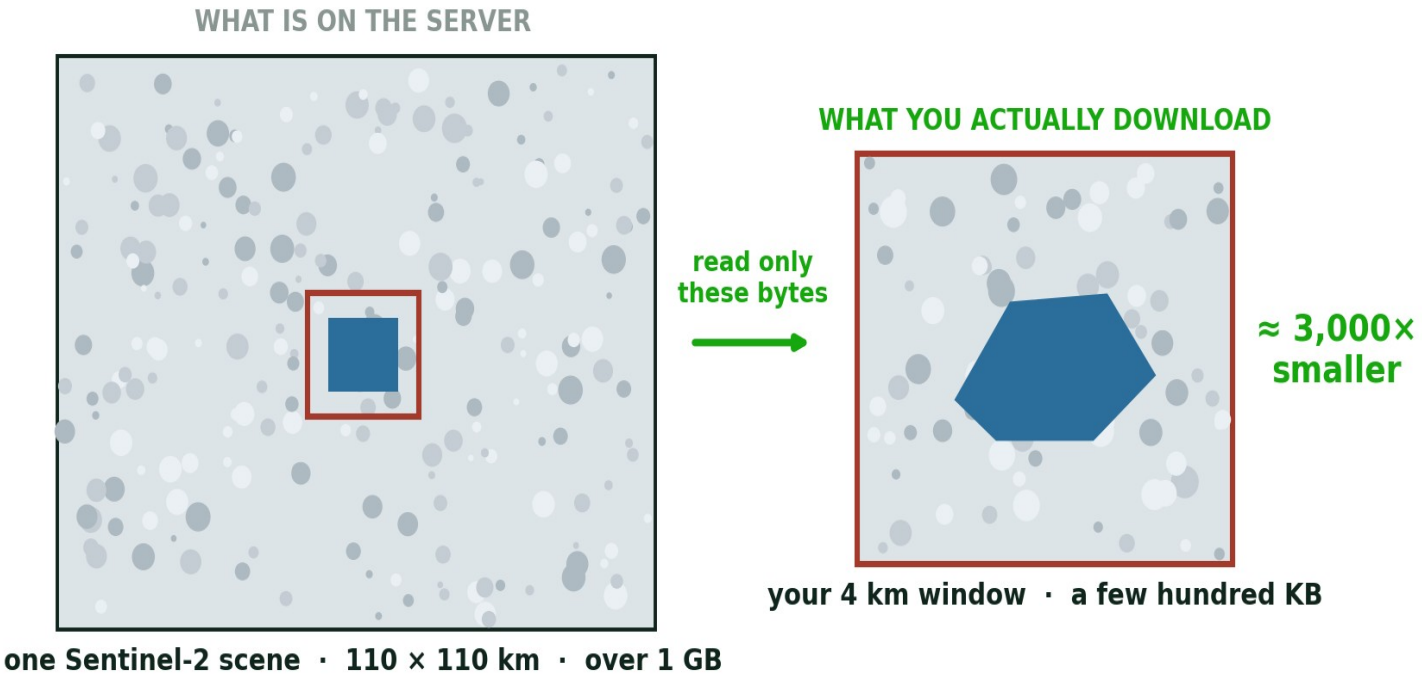




A camera waits for sunlight, and cloud stops it. Radar sends its own pulse, which passes straight through cloud. **It works at night, in monsoon, all year.**

8 How the data is actually read

We never download the scene



The mechanism

1. Ask a STAC catalogue for every scene over the lake in a date range.

2. Each file is a Cloud-Optimised GeoTIFF — internally tiled, with an index at the front.

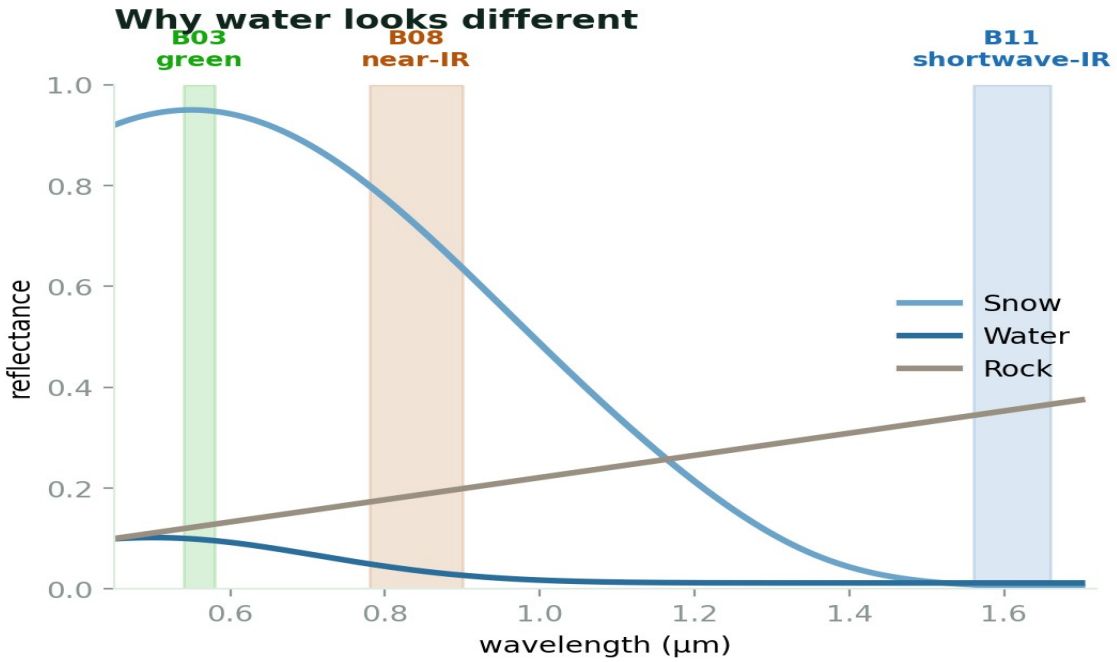
3. Request only the byte ranges covering our 4 km window.

4. Reduce to a few numbers in memory. Never store the raster.

The whole 8-year archive for 41 lakes is tens of megabytes, not terabytes — 1,145 small cached files. **It runs on a laptop.**

9 How a satellite decides a pixel is water

NDWI, MNDWI, and why one index is not enough



The two indices

```
NDWI  = (green − nearIR) / (green + nearIR)
MNDWI = (green − shortIR) / (green + shortIR)
```

surface	NDWI	MNDWI	
Water	+0.65	+0.78	high on both
Snow	+0.55	-0.72	high on NDWI, LOW on MNDWI
Rock	-0.28	-0.35	low on both

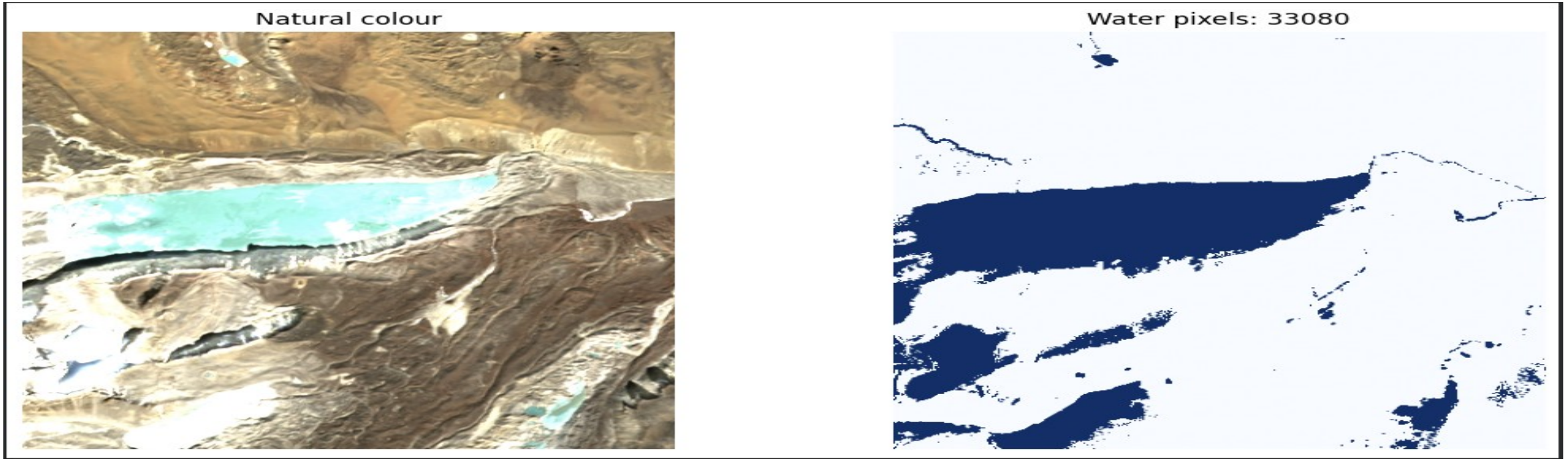
The trap

Snow passes the NDWI test. Only MNDWI rejects it — which is why both must be above 0.30, not just NDWI above 0.

Water absorbs infrared; snow does not — but snow reflects green just as strongly. **Snow passes the NDWI test and fails MNDWI, so both must agree.**

10 The trap, on a real Sentinel-2 scene

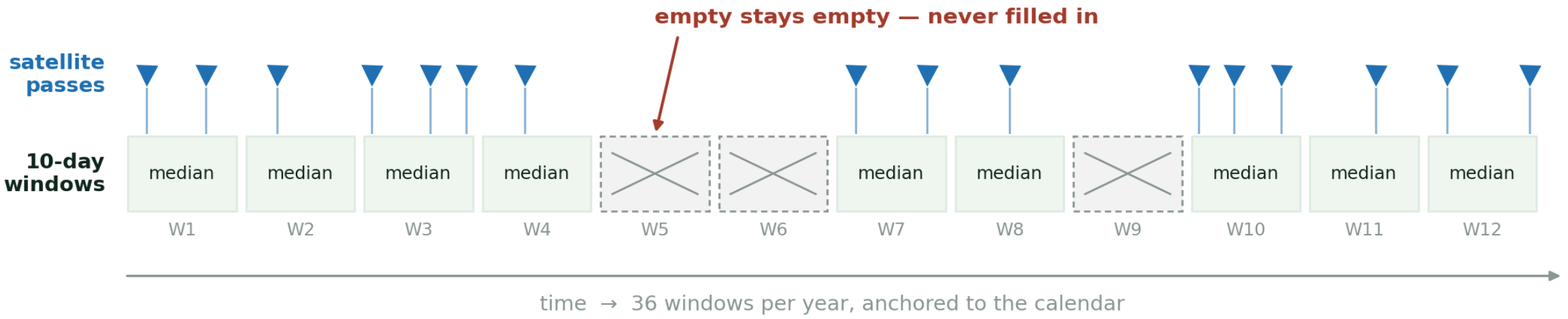
What the textbook threshold measures when the lake sits in permanent snow



Right panel: every pixel with NDWI > 0 — 33,080 of them, spilling far beyond the water onto snow and shaded slopes. **The fix is NDWI > 0.30 AND MNDWI > 0.30.**

11 Putting time on a fixed grid

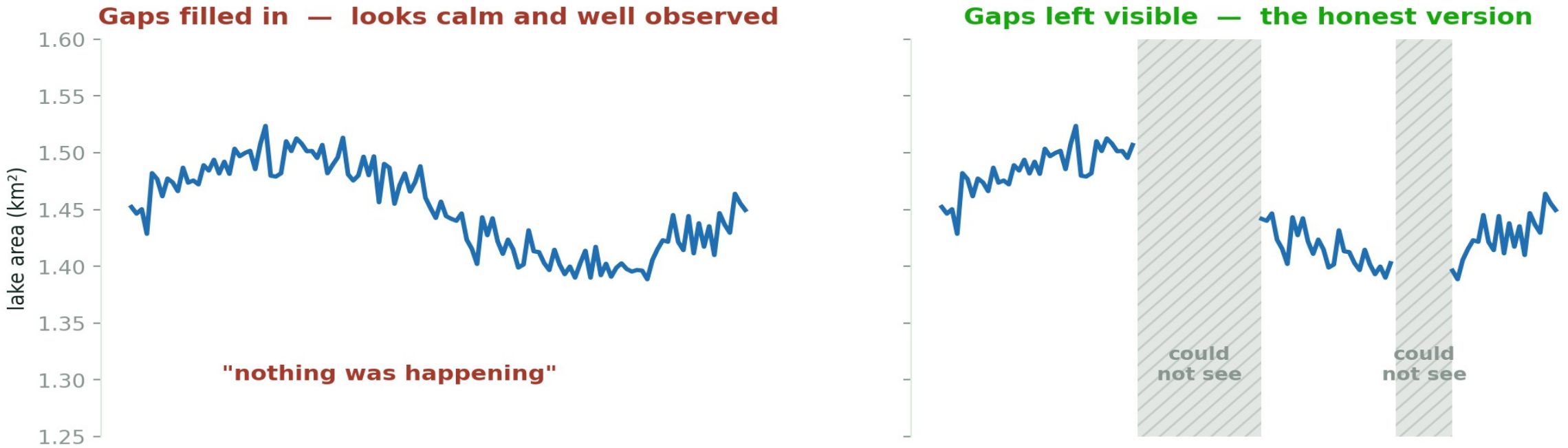
Different satellites, different days — one common calendar



36 windows a year, anchored to the calendar, identical for every lake and every sensor. **Windows with no usable observation stay empty — they are never filled in.**

12 Why gaps are never filled

The same data, two charts — one of them lies



A line drawn across a gap converts "we were blind for 40 days" into "nothing was happening". **Those are opposite claims about the safety of a lake.**

13 From pixels to five numbers

Each lake, each 10-day window, reduced to five signals plus an observability score

Signal	Units	In plain words	Conditioning
lake_area_km2	km^2	How big is the lake right now?	deseasonalised against its own causal climatology
velocity_trend	$m\ yr^{-1}$ per yr	Is the glacier speeding up?	already a difference
velocity_accel	$m\ yr^{-2}$	Is that speed-up accelerating?	already a second difference
thermal_index	°C-days	How hard is the melt season pushing?	deseasonalised
sar_backscatter_change	dB	Has the radar texture changed?	already a difference

Plus a sixth number that is NOT a feature: observability

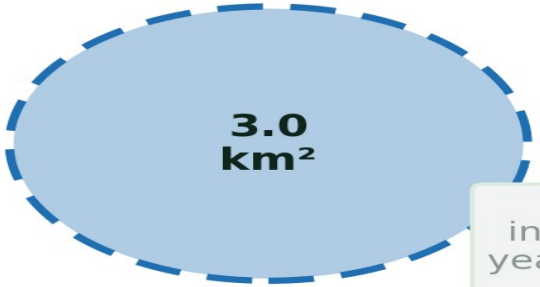
How much of this window we could actually see, weighted across sensors (0.40 / 0.30 / 0.15 / 0.15). It never enters the model — it decides whether the model runs at all. Below a floor of 0.30, the system abstains.

14 Why every lake gets its own normal

A single universal threshold would be wrong for every lake at once

The same absolute change means opposite things — so every lake is compared only to itself

A big lake



+0.1 km²

= 3% bigger

inside its ordinary year-to-year wobble

A small lake



+0.1 km²

= 33% bigger

an event

So 41 lakes means 41 separately fitted models. **The question is never "is this lake big?" but "is this lake unlike itself?" — which IS comparable across lakes.**

15 Deseasonalisation

Every shop sells more in December; every glacial lake grows in July



The seasonal normal for each of the 36 yearly slots is built from EARLIER YEARS ONLY — never from the future.

Compare this July against previous Julys, not against January. 36 separate normals per lake — and each built from EARLIER YEARS ONLY.

Measured price: a 9.8% warm-up cost.

16 Why there is no supervised model

The honest answer is a property of the world, not a modelling preference

Inside the study period and region there is exactly ONE confirmed moraine-dam failure.

South Lhonak, 4 October 2023. That is the entire positive class. n = 1.

What a classifier would do

Trained on one positive example it would memorise that lake's cloud pattern and sensor quirks, report near-perfect accuracy, and have learned nothing about failure.

What GEWS does instead

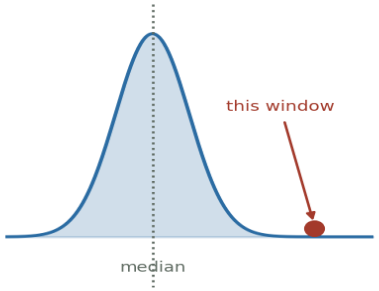
A one-class formulation. The model sees only ordinary behaviour, learns the shape of "normal", and scores how far outside it a window sits. No failures needed for training.

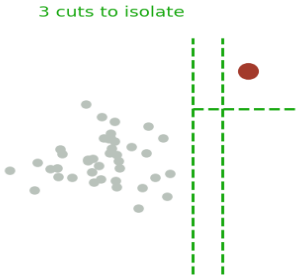
Which algorithms, precisely

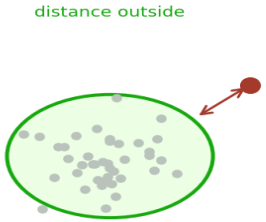
Four detectors exist — zscore (baseline), isolation_forest, one_class_svm, ridge_ar — but only ONE produces the results: isolation_forest, fitted 41 times, once per lake. The other three exist so the ablation can show the chosen one earns its complexity. With n = 1, no pooled precision or recall is computed, and the code refuses to compute one.

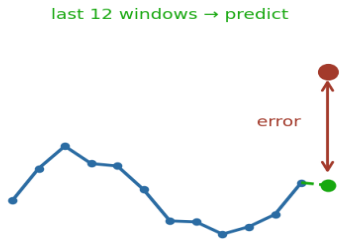
17 The five detectors

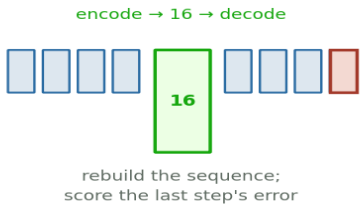
All one-class: they learn only what ordinary looks like, never what failure looks like











Robust z-score

`zscore`

How many MADs from this lake's median, worst signal wins. One signal at a time.

`median + MAD x 1.4826`

BASELINE · must be beaten

Isolation Forest

`isolation_forest`

Random cuts through the 5-D cloud. An odd window falls out in few cuts.

`300 trees · 256 samples`
`contamination 0.02`

PRODUCTION · this is the one

One-Class SVM

`one_class_svm`

Draw a boundary around normal; score how far outside a window lands.

`rbf · nu 0.02 · gamma scale`

ABLATION · better on slow drift

Ridge AR

`ridge_ar`

Predict this window from the previous 12; the anomaly is the prediction error.

`L = 12 windows · alpha 1.0`

ABLATION · time-aware

LSTM Autoencoder

`lstm_autoencoder`

Same idea, deep. Included so "would a neural net do better?" is measured.

`hidden 16 · 120 epochs`
`patience 15`

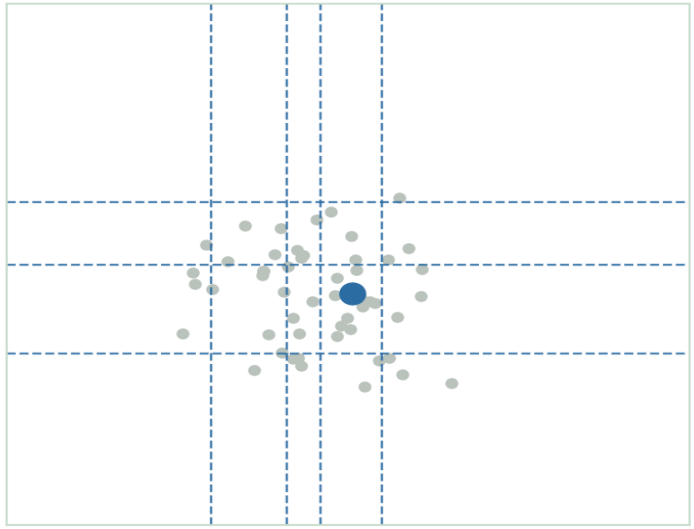
ABLATION ONLY · ~200 rows/lake

Only isolation_forest produces the published numbers. **The other four exist so the ablation can prove the chosen one earns its complexity** — including the LSTM, which with ~200 windows per lake has more parameters than data.

18 Inside the chosen model

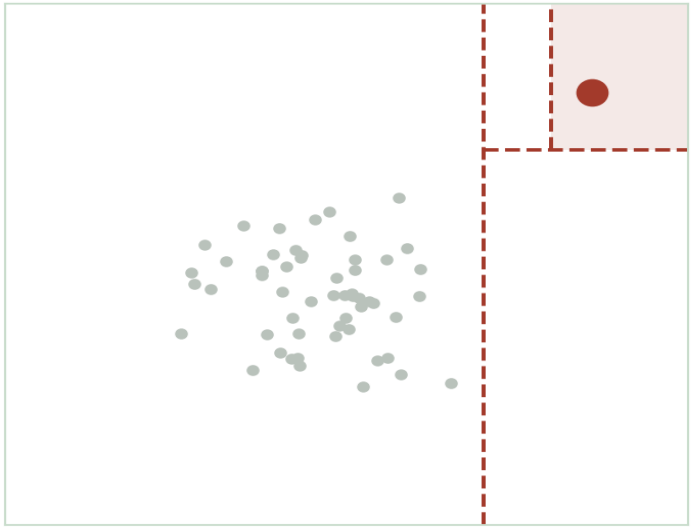
Isolation Forest — and how it names the reason, not just the score

An ordinary window



7 random cuts and it is still sitting with its neighbours

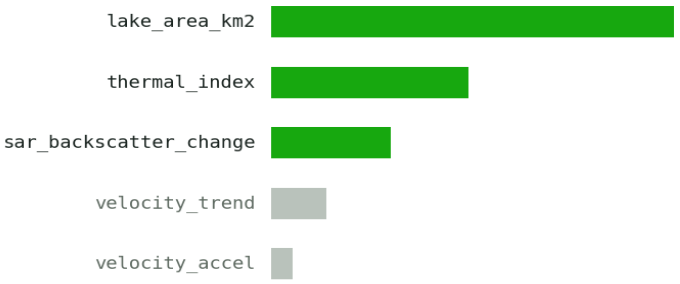
An unusual window



3 cuts and it is alone in a box. Short path = high anomaly score.

Which signal made it odd?

Leave one signal out, re-score, measure the drop



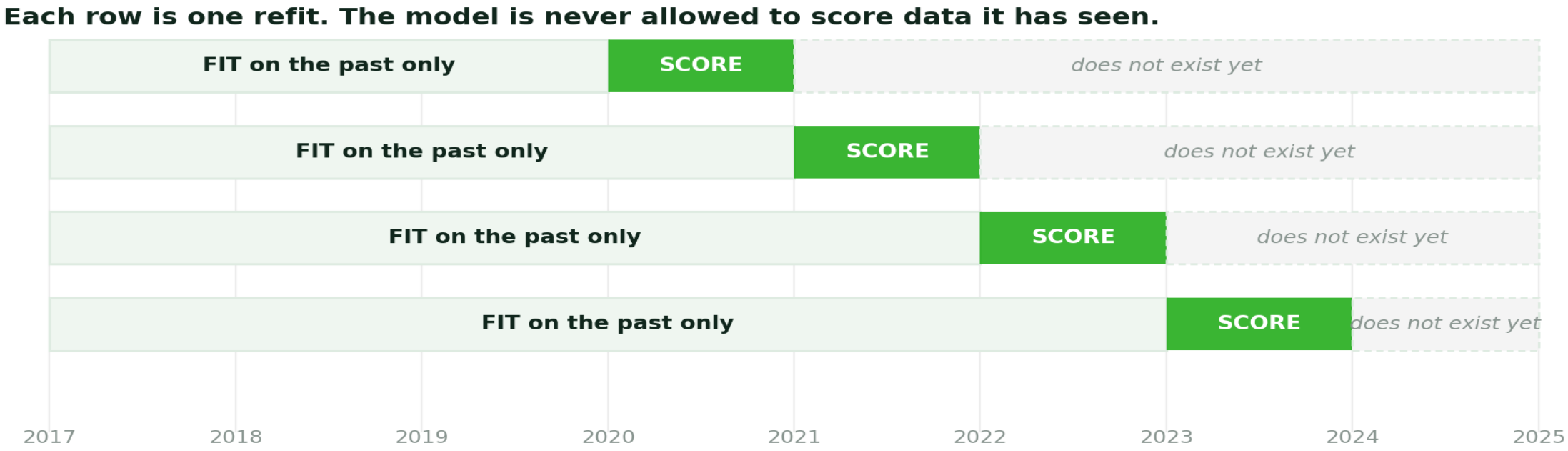
The top 3 become contributing_signals — what the officer reads as the reason.

```
n_contributing_signals: 3
```

Cut the 5-D cloud at random. An ordinary window keeps company; an odd one is alone after a few cuts. **The reason comes from leaving each signal out and measuring the drop.**

19 Walk-forward fitting

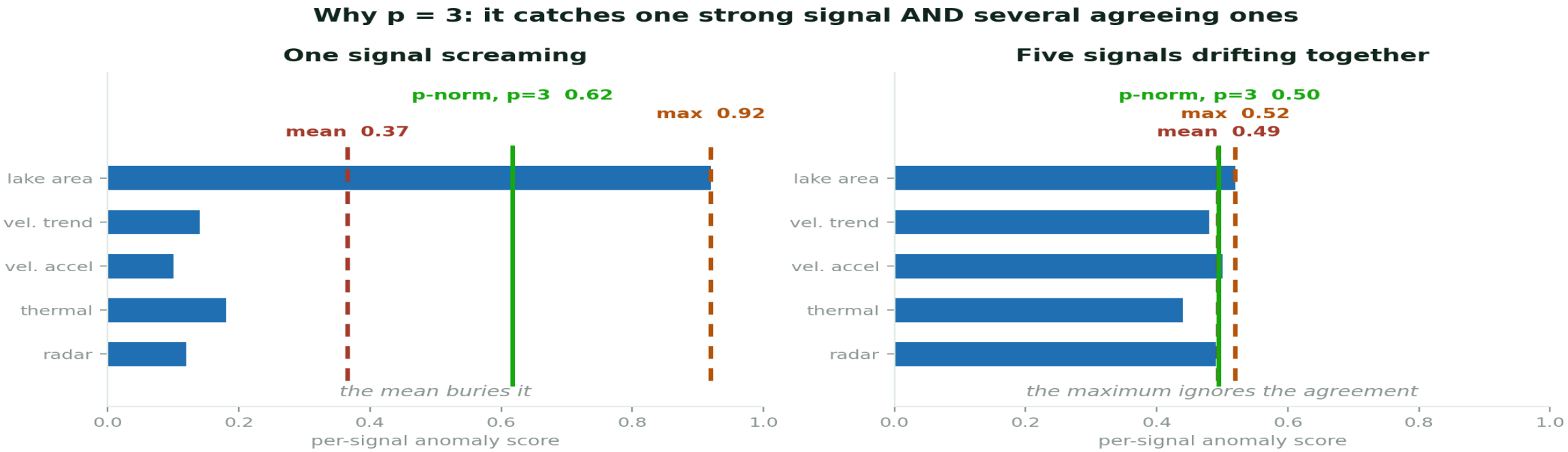
No model is ever allowed to score data it has already seen



Fit on the past, score forward, move the line, refit — every 36 windows. **Fit once on all eight years instead and results improve markedly. That improvement is exactly how much you would be fooling yourself by.**

20 Fusion — five scores into one

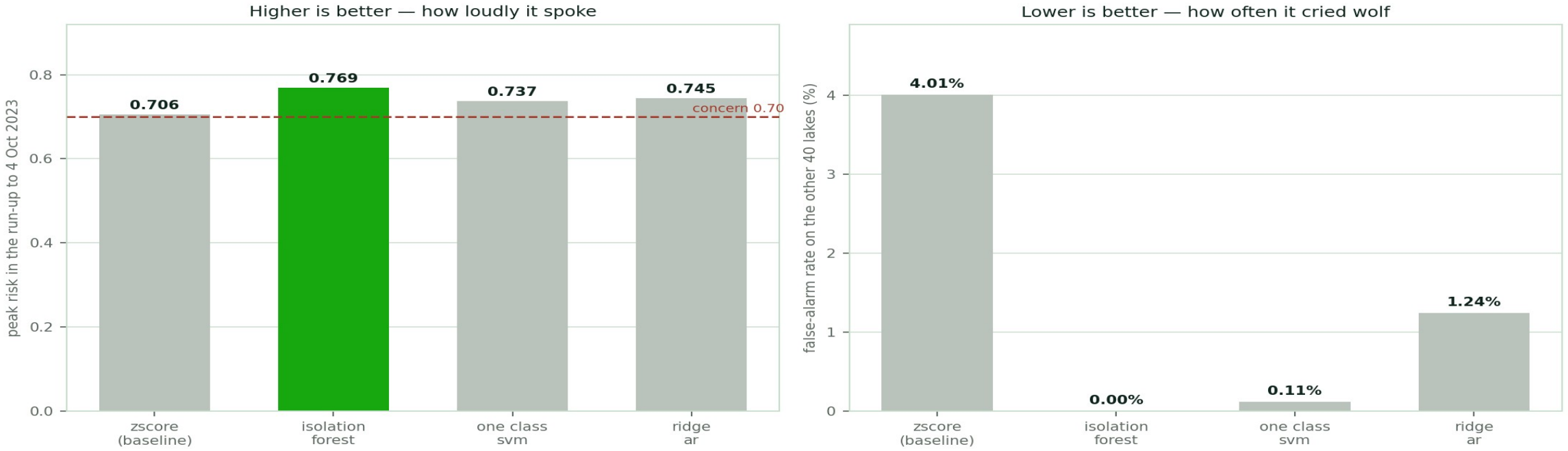
Weighted p-norm with p = 3



A mean buries one screaming signal; a maximum ignores five signals agreeing. **p = 3 rewards agreement without discarding a single strong signal.**

21 Why isolation_forest — measured, not chosen

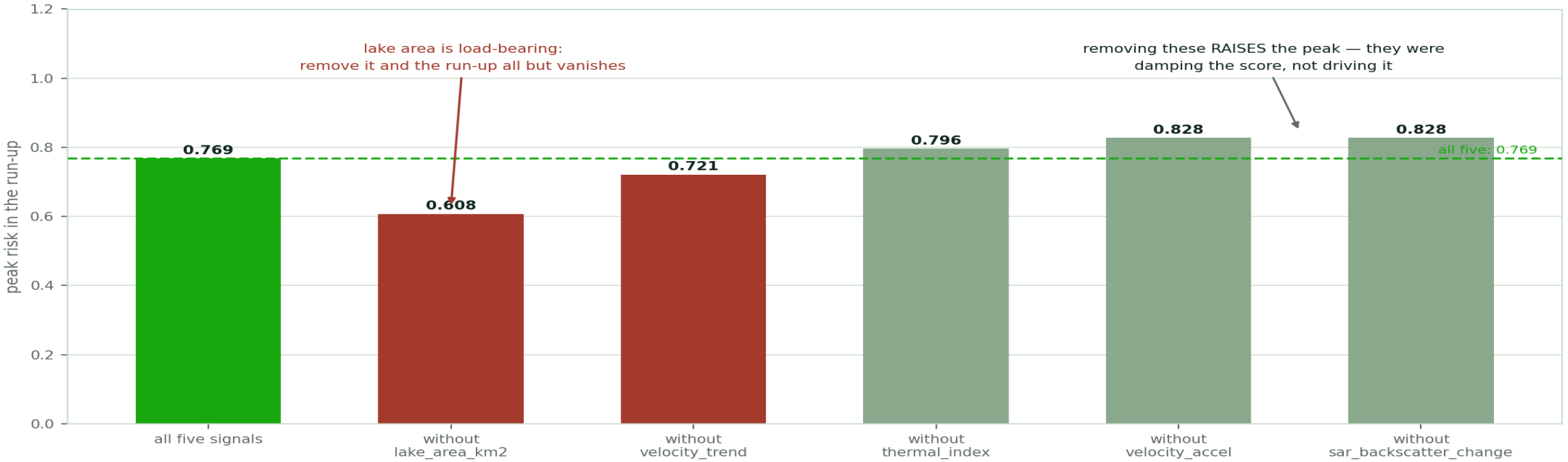
All four detectors were run on the same data; the table decided



The forest wins on both axes at once: the loudest run-up (0.769) and **zero false alarms across the other 40 lakes.** The simple z-score baseline nearly matches it on peak risk — 4% false alarms is what rules it out.

22 Which signals are load-bearing

Remove one at a time and re-run the whole pipeline



Lake area is the one that matters — remove it and the run-up all but vanishes. **Removing velocity_accel or the radar change RAISES the peak** — they were damping the score, not driving it.

23 How the risk ranking is produced

Five steps, in this order — and step 1 comes before anything is computed

1 Assessable?

Observability below 0.30, or fewer than 24 windows of history → NaN, ABSTAIN, reason. Stop.

2 Score per signal

That lake's own fitted model scores the window, one score per signal, each 0 to 1.

3 Fuse

Weighted p-norm, p = 3, over the available signals. Weights renormalise if one is missing.

THE FUSION STEP

$$risk = \left(\Sigma\, w_i \cdot s_i^3 \,/\, \Sigma\, w_i \right)^{(1/3)}$$

area 0.30 · vel_trend 0.20 · vel_accel 0.20
radar 0.20 · thermal 0.10

4 · Threshold 5 · Sort

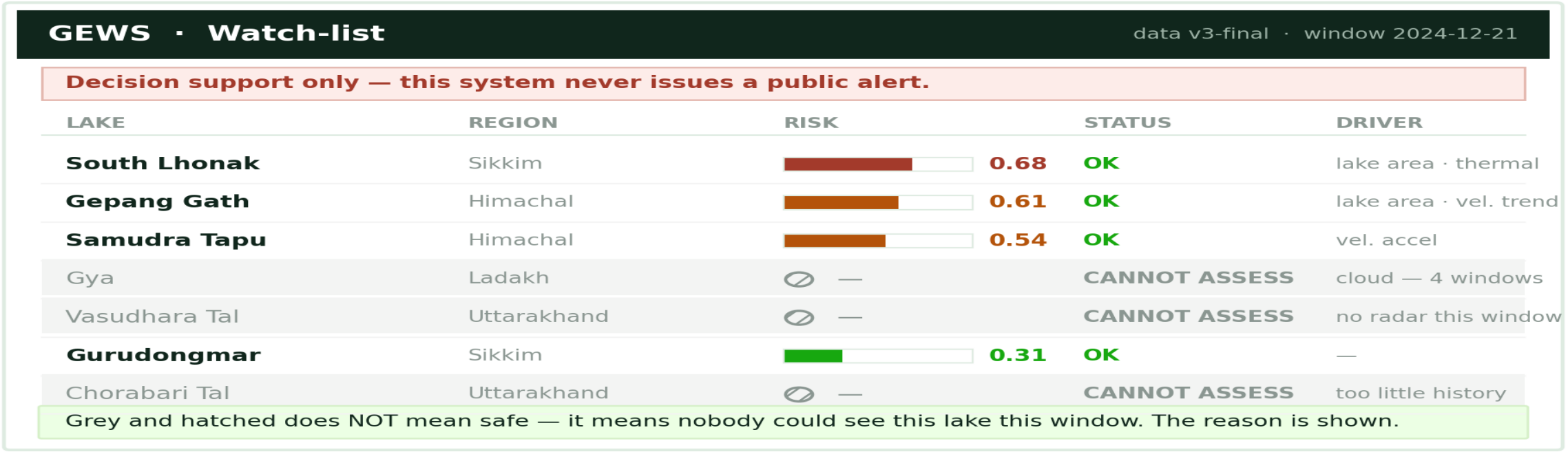
watch 0.50 · concern 0.70 · high 0.85, overridable per lake.

Then sort descending. Abstaining lakes are not ranked in with the numbered ones — you cannot sort a NaN, and must not sort it as zero.

This ranks unusualness, not danger.
A small harmless lake behaving oddly outranks a large dangerous lake behaving exactly as it always has. The fix is a separate hazard index, not a change to the ranking.

24 What the officer sees

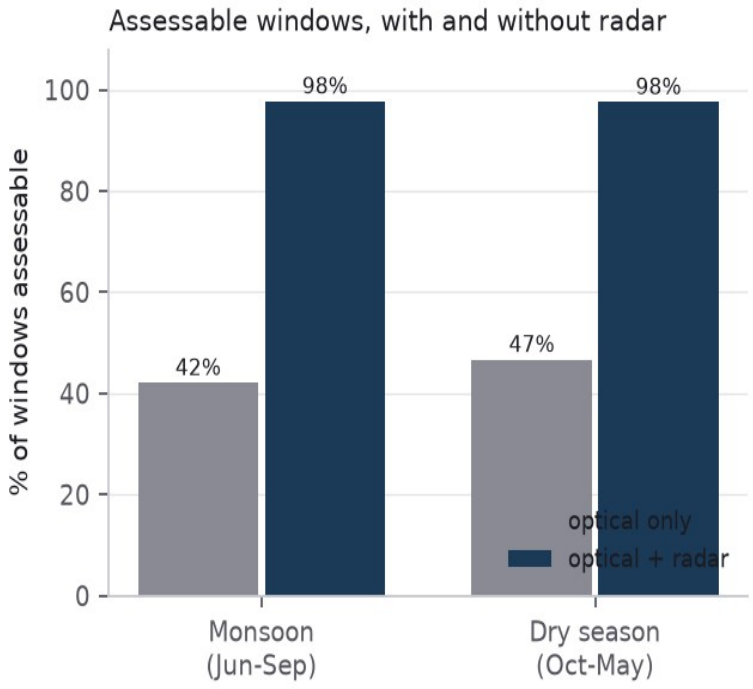
The rule was: understandable in five seconds

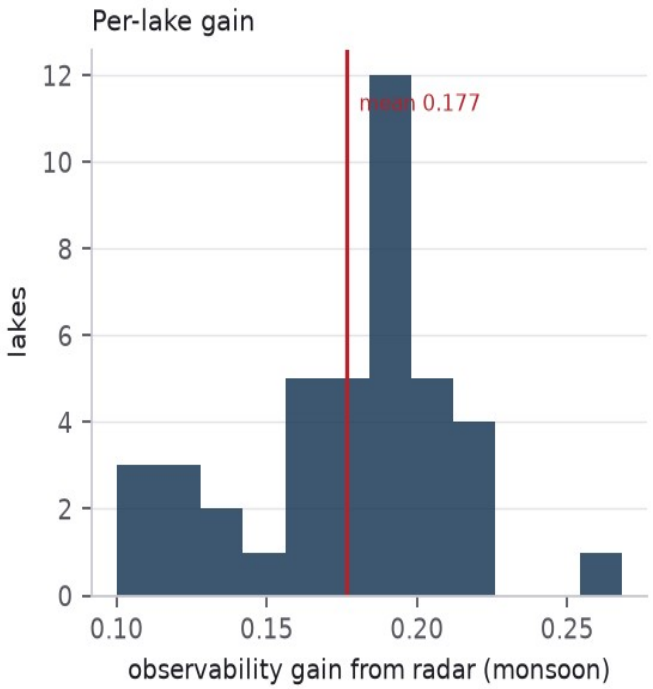


Grey and hatched does NOT mean safe — it means nobody could see this lake this window, and the reason is shown. **One button exports an A4 evidence PDF with a signature block.**

Result 1 · Radar rescues the monsoon

The strongest result, because it does not depend on any single event





What the numbers say

Monsoon windows assessable:

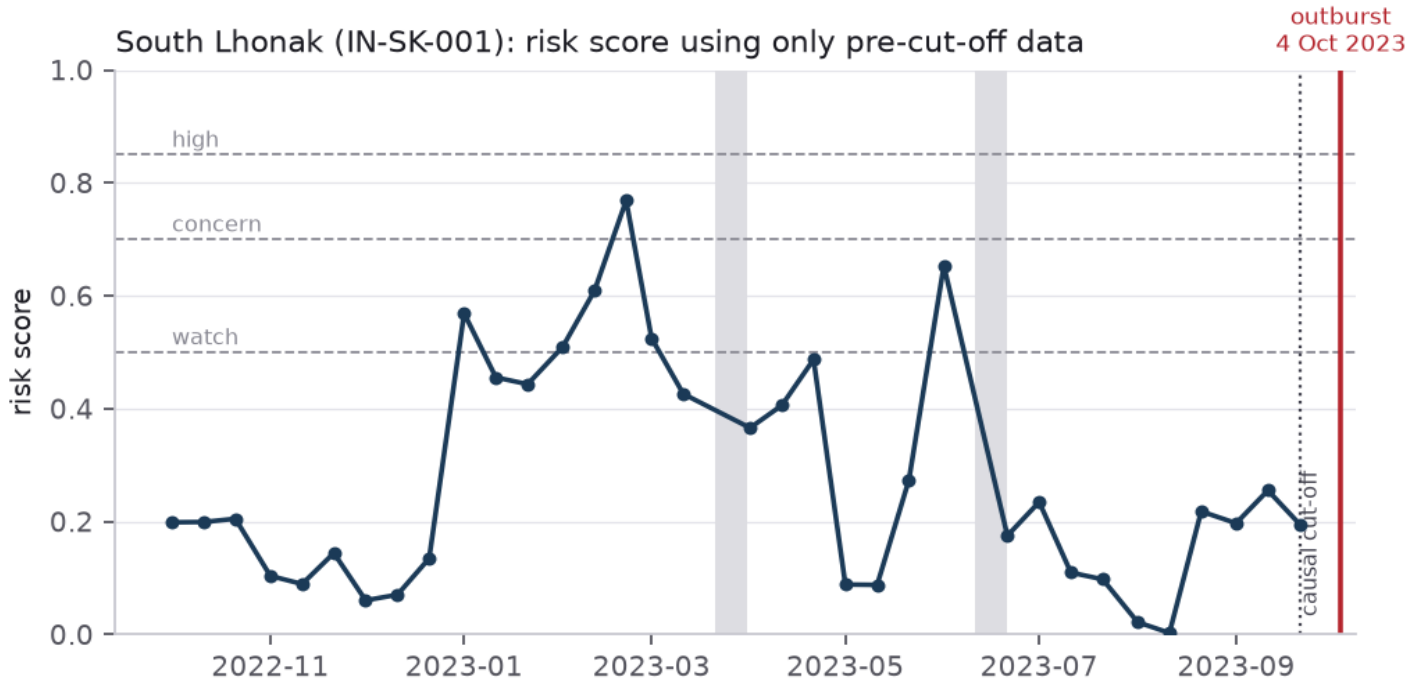
optical only	42.1%
optical + radar	97.7%

2,197 monsoon windows recovered of 3,936. Mean per-lake observability gain 0.177 (95% CI 0.174–0.179).

It rests on thousands of windows across 41 lakes over 8 years, not on one case. **Adding Sentinel-1 more than doubles how often these lakes can be assessed.**

26 Result 2 · The retrospective test is negative

Using only data from before 4 October 2023



What the run-up shows

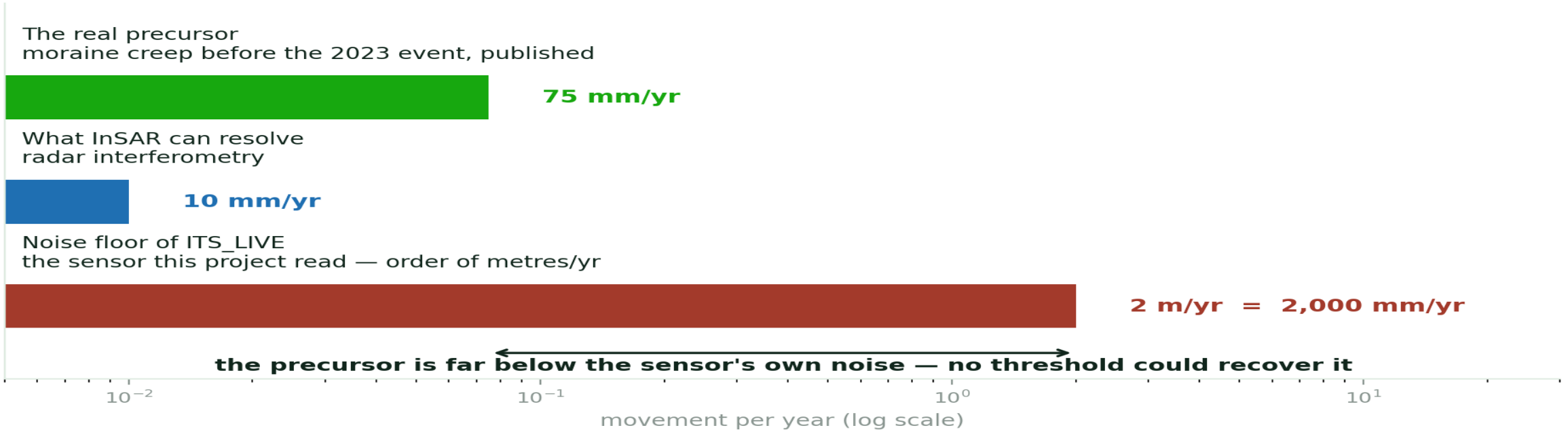
- high (0.85): never crossed
- concern (0.70): never crossed
- watch (0.50): crossed 245 days out, at a false-alarm burden of 24 alarm-days per lake-year

Risk peaks 7 months before the event, then falls. The direction of travel into the outburst is downward.

This null was not tuned away. Thresholds and fusion weights were frozen before the test ran — **moving any of them afterwards would produce a "success" that meant nothing.**

27 Why the miss is an instrument limit

The strongest scientific claim in the project



The documented precursor was moraine creep at 70–80 mm/yr (Yu et al. 2024, SBAS-InSAR). ITS_LIVE tracks features on 120 m pixels with a noise floor of metres per year. **No threshold could recover it — which turns "add InSAR" into a specification with a target sensitivity.**

28 Limits, and what comes next

Everything the system cannot do, stated before anyone asks

- The positive class has one member. No pooled precision or recall is computed, and the code refuses to compute one.
- Lead time is threshold-dependent and so is its cost. A lead-time number without its false-alarm rate is not a result.
- Blind for much of the monsoon even with radar. Worst measured blind spell: 130 consecutive days.
- Lake area is inferred from a water index over a fixed ROI, not surveyed. The change is the usable signal, not the absolute value.
- Slow monotonic growth is invisible — the model refits yearly, so a steadily growing lake stays "normal" to it.
- A precursor is not a cause. Drift without failure is common.

Next · InSAR

The answer the null result specifies. Coherence over the moraine face measures tens of mm/yr directly. 1–2 weeks.

Next · a hazard index

Absolute and slow-moving, beside the anomaly score: multi-year growth, freeboard, dam geometry, downstream exposure.

Next · widen the register

Three documented in-period events fell 12–85 km outside every ROI. The register covers the largest lakes, not the most dangerous.

29 Terminology

Every term used in this deck

Term	Plain meaning	Term	Plain meaning
GLOF	Glacial Lake Outburst Flood	Observability	How much of a window we could see. Below 0.30 → abstain
Moraine	Ridge of loose rock a glacier leaves. Here, the dam	Abstain	Refusing to score. Shown as "cannot assess", never a low risk
ROI	Region of interest — the 4 km square per lake	Walk-forward	Fit on the past, score forward — how it would have run at the time
NDWI / MNDWI	Water indices from colour bands. Snow fools the naive one	Deseasonalise	Compare this July with previous Julys, not with January
SAR / dB	Radar imaging; brightness in decibels	One-class	Learn only what normal looks like. Needs no failures
STAC / COG	Standards for finding scenes and reading part of one remotely	p-norm fusion	Combining five scores so agreement counts but a spike survives
Window	One 10-day step. 36 per year, same for every lake	Lead time	How far ahead an alarm crossed. Meaningless without its false-alarm rate
Feature	One of the five numbers the model reads	InSAR	Radar interferometry — millimetre ground movement. The sensor this needs

The project in seven lines

- Himalayan glacial lakes sit behind walls of loose rubble; when one fails the valley below has under two hours.
- Sensors are impractical at 5,000 m, so GEWS watches 41 lakes with four free satellites, 2017–2024.
- Each lake is reduced to five signals per 10-day window and compared only with its own past — 41 separate models.
- Where it cannot see it says "cannot assess" — NaN, never 0.0. That rule is enforced by tests, including the colours.
- Radar more than doubles monsoon assessability, 42% → 98%. Thousands of windows, not one event.
- On the one confirmed event it gave no warning at any operable threshold — and the reason is measured, not guessed.
- The precursor was 70–80 mm/yr; the sensor's noise floor was metres per year. That null specifies the next instrument.

Decision support only. Never a guarantee of safety. Where the system cannot see, it says so.

THANK YOU

Questions?

GEWS — Glacial Lake Outburst Flood Early-Warning System

Dept. of Artificial Intelligence & Machine Learning, DSCE

Data: ESA Copernicus · NASA · NRSC/ISRO. Code MIT-licensed. Total data cost: zero.

Decision support only · never a public alert